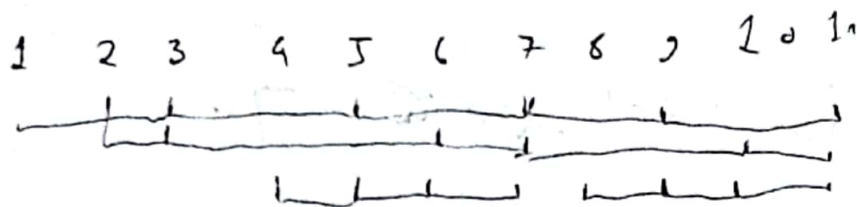


Ans. to Q no.



total valid codeword = $2^7 = 128$

~~Here $K=1$ i.e. for 1 bit~~

now $2^{11-7} = 2^4 = 16 > (1+11)$

So, $K=1$ i.e. for 1 bit error correction code is sufficient

For 2 bit error detection, it is also sufficient

Since every data bit contributes to at least 2 check bits (each set different). So, 1 bit error

can be easily checked.

So if from the set of contributions

3 - 1, 2

5 - 1, 4

6 - 2, 4

7 - 1, 2, 4

9 - 1, 8

10 - 2, 8

11, - 1, 2, 8

So, if two data bits are changed, as sets are different, at least one check bit will be inconsistent.

if ~~the~~ check bits are changed, obviously the odd parity ~~is~~ will not match. So, with 2 bit changes, valid codes cannot change valid. therefore, we can recognize two bit

errors with this code.

However note that, simultaneous K bit error correction and $K+1$ bit error detection is not possible.

For example if 1 and 3 bits flip, we cannot distinguish

1 bit flip or 2 bit flip.